Opinion Mining: Improving Opinion Classification Using a Combination of Supervised and Unsupervised Approaches

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Abstract

As is well know, there is a continuing growth of personal opinions on the web. Effective use of this information would be beneficial to both consumers and manufacturers. This paper presents a novel and unique opinion classification method by combining supervised and unsupervised approaches. Based on the merits of these approaches and the characteristics of opinions, we first separate an opinion into attribute expressions and reputation expressions using the part-of-speech of the words. Then feature classification is performed by a supervised approach (Naive Bayes) for the attribute expressions. P/N (Positive/Negative) classification of the reputation expressions is performed by an unsupervised approach (SO-PMI: Semantic Orientation Using Pointwise Mutual Information). The unsupervised method has previously been successfully used on English. We improved it for processing Japanese semantic orientation. Our experimental results show the effectiveness of the proposed method.

1 Introduction

With the rapid expansion of the Internet, the web has become an excellent source for gathering consumer opinions. There are now numerous Web sites containing such opinions, e.g. customer reviews of products, forums, discussion groups, and blogs. Here, we use the term Weblog for these sites. Obviously, there is a lot of valuable information in these opinions. Knowing the reputations of your own and/or competitors' products is very important for marketing and customer relationship management. However, opinion information in the Weblog makes up an enormous amount of unstructured and/or semi-structured data. These data are factual, but in free format and mostly textual. It is very hard and costly to collect and analyze them manually. Opinion mining integrates some relevant techniques so as to handle such unstructured data effectively.

Related work has mostly studied how to classify the opinions into bipolar orientation i.e. either positive or negative opinion (Chaovalit, 2005; Turney, 2002; Kobayashi, 2003). These research results make it possible for users to reduce the time spent on reading threads of text and focusing more on analyzing summarized information. These works usually use either a supervised classification approach (Gamon, 2005; Pang, 2002) or an unsupervised classification approach (Popescu, 2005; Turney, 2002), or give a comparison of these approaches by classifying the same data (Chaovalit, 2005).

This study proposes a novel and unique opinion classification method by combining supervised and unsupervised approaches. We demonstrate the advantages of this approache by implementing different classification tasks in opinion mining. Based on the grammatical structure of Japanese, an opinion generally consists of an attribute and a reputation portion. We present an opinion separation method that separates one opinion phrase into attribute expressions and reputation expressions. Then these expressions are classified by supervised and unsupervised approaches. Both supervised and unsupervised approaches have their pros and cons. Supervised approaches are likely to provide more accurate

classification results but need a training corpus. Unsupervised approaches on the other hand, adapt to semantic orientation like to measure how far a word is inclined towards positive and negative.

Since it is easy to create a feature corpus, we perform feature classification which classifies attribute expression into Price, Design and Function features using a supervised approach, Naive Bayes. Since semantic orientation on the other hand is used to analyze the orientation of a word (positive or negative), we perform P/N classification which classifies reputation expressions into positive and negative meaning using an unsupervised approach, SO-PMI: Semantic Orientation Using Pointwise Mutual Information (Turney, 2002). To the best of our knowledge, such a combination has never been reported before.

The remainder of this paper is organized as follows: Section 2 provides the background of this study. Section 3 describes the proposed classification method in detail. In section 4, we evaluate and discuss the experimental results. Section 5 gives concluding remarks.

2 Background

2.1 Mining Technology

Text mining is a cross between Information Retrieval (IR), Data Mining (DM), Artificial Intelligence (AI), and Natural Language Processing (NLP). It is also known as intelligent text analysis, text data mining or Knowledge Discovery in Text (KDT), and generally refers to the process of extracting information and knowledge, facts and meaningful relationships, from text documents. Since most company data (over 80%) is stored as text, text mining is believed to have a high commercial potential value (Yasuda, 2005; Gamon, 2005).

Opinion mining is a sub discipline of text mining. The purpose of opinion mining is to mine reviews of various products by classifying them into positive or negative meaning from the 'subjective' terms contained in the document. Automated opinion mining can provide quick search and analysis results to both consumers and manufacturers (Chaovalit, 2005).

Web mining is the application of text mining techniques to extract valuable knowledge from various types of data on the Web, including logs, cookies, links, and contents. The current research area of Web mining includes 1) Web content mining, 2) Web structure mining, 3) Web usage mining (Kosala, 2000). In this study, Web content mining, or more specifically the mining of the Weblog, is studied.

2.2 Text Classification

Text classification is one of the core problems in Text Mining (Yasuda, 2005). It is a technique for dividing texts into two or more categories. For example, it is used for automatically assigning news articles to categories, such as politics, economy, and sports, based on the contents. There are many applications for classifying texts, for instance, analyzing the results of questionnaires, product opinions on Internet bulletin boards, or classifying e-mail as spam mail or regular mail.

Text classification methods can be divided into two types: supervised classification approaches and unsupervised classification approaches. Supervised classification is a machine learning technique for creating a function from training data. The training data consist of pairs of input objects and desired outputs. There exist many supervised learning methods, for example Support Vector Machines (SVM), Naive Bayes classifiers etc. Unsupervised classification on the other hand does not require a prior training corpus. For example, the semantic orientation approach measures how far a word is inclined towards positive and negative to classify a review as recommended or not recommended (Chaovalit, 2005).

Supervised classification is likely to provide more accurate classification result than unsupervised classification. It is tuned to the training corpus, though, and thus needs retraining if it is to be applied in a new domain.

2.3 Related Works

The problem of "opinion mining" has seen increasing attention over the last three years, from (Chaovalit, 2005; Turney, 2002; Kobayashi, 2003) and many others. Our work is closely related with the following studies.

Turney (2002) works on opinion classification using word semantic orientation. He presents an unsupervised learning algorithm called SO-PMI for classifying reviews as recommended or not recom-

mended. The classification of a review is based on the average semantic orientation of the phrases in the review that contain adjectives or adverbs.

Kobayashi et al. (2003) define the main portions of an opinion as <object, attribute, opinion>. They propose a technique for efficiently building an object name dictionary, an attribute expression dictionary, and an opinion word dictionary for any object domain. Chaovalit's study (2005) gives a comparison between a supervised approach (machine learning) and an unsupervised approach (semantic orientation) on movie opinion mining.

3 Proposed Classification Method

In this section, an improved opinion classification method is presented in detail. There are two kinds of classification here: feature classification and P/N classification. The purpose of feature classification is to know what an opinion evaluates based on the attribute of the opinion. P/N Classification aims to understand what the opinion expresses about the feature. Opinion Separation is a preprocessing step for the classification.

3.1 Opinion Separation

Based on the grammatical structure of Japanese and the results in related works (Kobayashi, 2003; Taku, 2002), an opinion generally consists of an attribute expression and a reputation expression. The attributes of products are almost always nouns, and the reputations are almost always an adjective or verb. For example, "この色は素敵です (This color is lovely)" . "色" is noun that means "color". "素敵" is adjective that means "lovely".

Therefore, for classifying opinions efficiently, we propose an Opinion Separation Method presented in Figure 1. The opinion phrases are separated into attribute expressions and reputation expressions according to the POS (part-of-speech) of the words. If the phrase is a noun phrase, it would be classified as an attribute expression. If the phrase consists of an adjective or verb, it would be classified as a reputation expression. These expressions are then classified into feature and P/N categories respectively, which are described in detail in the next sections.

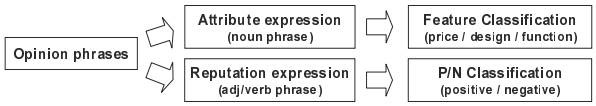


Figure 1: Opinion separation

3.2 Feature Classification

Feature classification aims to classify attribute expressions into Price, Design or Function features. After this classification, we can know what an opinion evaluates. In this paper, we limited the feature categories to only these three main and common features.

Through analyzing various product review information on the Weblog, we discovered the following facts. Firstly, attribute expressions are almost always nouns. Secondly, different categories of products have similar attributes (such as " cost ", " size " etc). Thirdly, the number of this kind of expressions is limited. A sample of attribute expressions is shown in Table 1. This investigation indicated that it would be easy to create a training corpus. Creating a classifier would not take much time and reasonably accurate classification results could be expected. Therefore, we decide to use a supervised approach to perform this classification.

Table 1: Sample of product attribute expression

Feature	Attribute expression
Price	値段(price), 価格(cost), 料金(charge), 安値(low price). . .
Design	サイズ(size), デザイン(design), 幅(width), キー(key). . .
Function	再生(play), 辞書(dictionary), 発音(pronunciation), 機能(function). . .

The Naive Bayes algorithm is one of the oldest and most well-known supervised approaches (McCallum, 1998). To make the probability based classification computation feasible, Naive Bayes assume independence among the variables, in a naive fashion. For real problems this assumption is certainly a simplistic one. However, despite this objection, Naive Bayes works rather well when tested on actual databases, particularly when combined with some feature selection procedure.

3.2.1 Basic theory of Naive Bayes

Given a database of cases consisting of a set of predictor variables X_1, X_2, $\cdots$, X_n and a special variable Y whose value must be predicted, each of the N cases being of the form

$$Case_i : \\ X_1 = x_{1i}, X_2 = x_{2i}, \cdots, X_n = x_{ni}, Y = y_j \quad (1)$$

and being, without loss of generality, $Y = y_1$, $Y = y_2, \cdots, Y = y_m$ the m possible values the class variable can take (the m categories considered in the classification problem), theoretically, Bayes' rule minimizes error by selecting the class y_j with the largest posterior probability for a given example **X** of the form $X = < X_1 = x_1, X_2 = x_2, \cdots, X_n = x_n >$, as indicated below:

$$P(Y = y_j|X) = \frac{P(Y = y_j)P(\mathbf{X}|Y = y_j)}{P(\mathbf{X})} \quad (2)$$

Since **X** is a composition of n discrete values, one can expand this expression to:

$$P(Y = y_j|X_1 = x_1, \cdots, X_n = x_n) = A \\ A = \frac{P(Y=y_j)P(X_1=x_1,\cdots,X_n=x_n|Y=y_j)}{P(X_1=x_{1i},\cdots,X_n=x_n)} \quad (3)$$

where $P(X_1 = x_1, \cdots, X_n = x_n|Y = y_j)$ is the conditional probability of the instance **X** given the class y_j; $P(Y = y_j)$ is the a priori probability that one will observe class y_j; P **(X)** is the prior probability of observing the instance **X**. All these parameters are estimated from the training set.

3.3 P/N Classification

P/N classification means to classify the phrases into positive or negative meanings using SO (Semantic Orientation) values. Its purpose is to understand what the opinion expresses, recommended or not recommended. We improved the SO-PMI approach for performing Japanese semantic orientation.

3.3.1 SO-PMI

Following Turney's study (2002), the SO-PMI algorithm is employed to estimate the semantic orientation of a phrase by measuring the similarity of pairs of words or phrase. The SO of a phrase is calculated here using the following formula:

$$SO(phrase) = PMI(phrase, \text{“}excellent\text{”}) \\ - PMI(phrase, \text{“}poor\text{”}) \quad (4)$$

$$PMI(word_1, word_2) = log_2 \left[\frac{p(word_1 \& word_2)}{p(word_1)p(word_2)} \right] \quad (5)$$

The reference words "excellent" and "poor" are used, so SO is positive when a phrase is more strongly associated with "excellent" and negative when a phrase is more strongly associated with "poor".

Let hits(query) be the number of hits returned when using a search engine, the following estimate of SO can be derived from Formula (4) and (5) with some minor algebraic manipulation.

$$SO(phrase) = log_2 [B] \\ B = \frac{hits(phrase\ NEAR \text{“}excellent\text{”}) * hits(\text{“}poor\text{”})}{hits(phrase\ NEAR \text{“}poor\text{”}) * hits(\text{“}excellent\text{”})} \quad (6)$$

3.3.2 Improvement of SO-PMI

On the basis of the characteristics of the Japanese language, we improved the SO formula to the one shown in Formula (7).

$$SO(phrase) = log_2 [C] + threshold \\ C = \frac{hits(phrase\ AND\ good_basic) * hits(bad_basic)}{hits(phrase\ AND\ bad_basic) * hits(good_basic)} \quad (7)$$

There are three main points about the improvement:

- Search Engine:

  We used the Google search engine[1] to get the hits(query). Though Google dose not have a NEAR operator like AltaVista which was used by Turney (2002), Google gets better results than AltaVista on searches in the Japanese Weblog. Google searches through more sites and faster, delivering the most relevant results. We chose Google and replaced the NEAR operator with the AND operator in the SO formula.

- Reference Words:

  In Japanese, especially in the Japanese Weblog, there are many different words for "excellent" and "poor" that are all highly frequent. Therefore, we replaced the reference words "excellent" and "poor" with "$good_basic$" and "bad_basic". "$good_basic$" means the basic popular words for positive opinions in Japanese. In contrast, "bad_basic" are the

[1] http://www.google.co.jp/

basic popular words for negative opinions in Japanese.

These words and how frequent they are is shown in **6.1**. The number of negative expressions is less than the number of positive expressions. We chose the positive words whose ratio of the whole Weblog pages is more than 5% as "$good_basic$" reference words, and negative words whose ratio of the whole Weblog pages is more than 3% as "bad_basic" reference words, shown in Table 2. The thresholds 5% and 3% were selected based on observations of the reputation words that people usually use to judge the quality of products in the Japanese Weblog. These words always have a comparatively steady ratio above 5% (positive) and 3% (negative). We found the same result when we investigated the same words on 2006.09.05 (shown in **6.1**) and 2006.07.19 (omitted).

Table 2: Reference words

	Reference Words
"good_basic"	いい｜好き｜良い｜大好き｜楽しい｜良かった｜嬉しい｜素敵｜面白い｜魅力｜良く｜欲しい｜よい｜満足｜素晴しい｜おもしろい｜うれしい *(いい: good / 好き: like / 良い: good / 大好き: favorite / 楽しい: delightful / 良かった: good / 嬉しい: happy / 素敵: lovely / 面白い: interesting / 良く: good / 魅力: charm / 欲しい: want / よい: good /満足: satisfaction / 素晴しい: wonderful / おもしろい: interesting / うれしい: happy)*
"bad_basic"	あまり｜悪い｜難しい｜嫌｜不安｜嫌い｜怖い｜苦手｜あんまり｜不良｜まずい｜悪く｜不満｜最悪｜危険｜だめ｜悲しい｜辛い｜不快 *(あまり: not good / 悪い: bad / 難しい: hard / 嫌: dislike / 不安: worry / 嫌い: dislike / 怖い: fearful / 苦手: hard / あんまり: not good / 不良: bad / まずい: bad / 悪く: bad / 不満: dissatisfaction / 最悪: worst / 危険: risk / だめ: useless / 悲しい: sad / 辛い: painful / 不快: dissatisfaction)*

- Threshold:

  In a preliminary experiment, we used zero as a threshold to separate positive and negative meanings. The result of this trial on a small sample data was extremely unsatisfying. Both positive and negative phrases seem to lean towards the negative end. Therefore, we decided to setup our own baseline to separate positive SO from negative SO. The threshold was adjusted until a better result was obtained. The threshold used was:

$$threshold = log_2\left[\frac{hits(bad_basic)}{hits(good_basic)}\right] * (-0.8) \quad (8)$$

4 Experimental Evaluation

4.1 Test Data

Table 3 shows the five types of IC Dictionary and five types of MP3 player that were used in the experiments. We extracted the opinion information of these products from the Japanese Weblog by these steps (Wang, 2006):

1. Information Search: Using the Google Search Engine to collect all the available information on the Internet. The search range is restricted by URL type (e.g. blog, bbs, review).

2. Weblog Content Extraction: Extracting the Weblog content according to the URL type for each of the collected pages.

3. Weblog Content Analysis: Using a well known dependency structure analyzer for Japanese called Cabocha (Cabocha, 2005) to analyze the Weblog content.

Table 3: Product type list for the test

Product Category	Product Type
IC Dictionary	T1: CASIO XD-ST6200
	T2: CASIO XD-GT6800
	T3: SHARP Papyrus PW-V8100
	T4: SHARP Papyrus PW-A8410
	T5: SEIKO SR-E10000
MP3	T6: SONY NW-E005
	T7: SONY NW-A1000
	T8: APPLE iPod MA002J/A
	T9: APPLE iPod nano MA005J/A
	T10: PANASONIC D-Snap Audio SV-SD750V-A

4.2 Feature Classification

As mentioned in **3.2**, we used a supervised approach (Naive Bayes) to perform feature classification. In this section, we discuss the training corpus and the results of the experiments.

4.2.1 Training Corpus

For a supervised approach, the most important thing is to train the classifier on an ideal corpus. Classification results depend on the quality of the training corpus. Since there are only three feature categories (Price, Design and Function), and the number of attribute expressions is limited, we decided to build the training corpus manually.

The data source is the Weblog content on the products T1, T2, T3, T4, T6, T7, T8 and T9 shown in Table 3. T5 and T10 are used for testing. By analyzing the Weblog information of these products, we created a corpus that has 30 attribute expressions of Price (F1), 80 attribute expressions of Design (F2) and 230 attribute expressions of Function (F3). A sample of attribute expressions is shown in Table 1.

4.2.2 Experimental Results

To verify the effectiveness of opinion separation, we compared the precision of "Before Separation" and "After Separation" by classifying the same attribute expression phrases of T5 and T10 into F1, F2 and F3 features. Table 4 shows the comparative results. PF_i means the precision of F_i. EP means the entire precision of feature classification, calculated by PF_i and w_i (the ratio of F_i) in Formula (9). w_1 is 0.1, w_2 is 0.15 and w_3 is 0.75.

Table 4: Experiment of Feature Classification

	PF1(%)	PF2(%)	PF3(%)	EP(%)
Before Separation	39.66	40.90	49.41	47.16
After Separation	87.50	85.71	56.79	64.20

PF1: [the precision of F1(Price)] FP2: [the precision of F2(Design)]

PF3: [the precision of F3(Function)] EP: [the Entire Precision)]

NF: [the number of feature(F1, F2, F3)]

$$EP = \sum_{i=1}^{3} w_i * PFi \quad w_i : NF_i / (\sum_{i=1}^{3} NF_i) \qquad (9)$$

After opinion separation we achieved higher precision, especially for the features F1 and F2. Since the number of examples of F1 and F2 was very much lower than the number for F3, the training corpus for these two features was more focused and the precision is higher than for F3. Over 70% of all feature expressions were of type F3. A lot of general words were classified into the F3 category. General words include for instance query expressions. There are a lot of general words and chats in the Weblog mixed with the opinion expression. Hence, how to distinguish this noise from opinion information is an important topic for future study.

4.3 P/N classification

4.3.1 Negated expressions

There are a lot of negated expressions in the Japanese reputation expressions, e.g. "not bad" used when meaning something positive. To evaluate the validity of the SO-PMI approach for managing negated expressions, we tested the system on some negated expressions listed in **6.2**. The results showed that the system can judge the negated expressions fairly well, with 21/26 (81%) examples classified correctly.

4.3.2 Experimental Results

As mentioned in **3.3.2**, we used an improved SO-PMI like Formula (7) to perform P/N classification. In a preliminary experiment, the threshold was set to zero. Both positive and negative precision was very unsatisfying. Possibly, in the time since Turney's study factors in calculating the SO have changed. It is also possible that different language sources give different results. Therefore, we adjusted the threshold based on the SO of reference words to the one in Formula (8).

Table 5: Precision of P/N Classification

	Threshold	Product Category	PP(%)	NP(%)	AP(%)
Preliminary	0	IC Dictionary	20.59	30.61	25.60
experiment		MP3	22.13	35.68	28.91
After	0.78	IC Dictionary	60.00	43.66	51.83
improvement		MP3	64.29	58.17	61.23

PP: [Positive Precision] NP: [Negative Precision] AP: [Average Precision]

After this improvement, we achieved better precision, which is shown in Table 5. The precision of negative opinions was still poor. The main cause is general words, there are a lot of general (neither positive nor negative) words in the negative classification results. There was no clear difference between the precision of the IC Dictionary and the MP3 player categories.

5 Conclusions

This study proposed an original opinion classification method using both supervised and unsupervised approaches to do Japanese opinion mining. We have presented in detail the feature classification of attribute expressions by a supervised approach (Naive Bayes), and the P/N classification of reputation expressions by an unsupervised approach (SO-PMI). The key to integrating these two approaches is the opinion separation method, which is based on the grammatical structure and other characteristics of opinion expressions in Japanese. We used real opinion information from the Weblog to empirically demonstrate that the proposed method is able to classify the opinions more precisely than simpler approaches. The precision of the feature classification was improved from 47% to 64%. The precision of P/N classification was improved from under 30% to 60%. Since there is a lot of noise in the Weblog, the precision was still not very high. However, we believe that with some further study of how to deal with the noise, our method could be improved even more and successfully be applied to mine opinions of various products from the Weblog.

In the future, we plan to proceed in mainly three directions. First, for the feature classification, we will add more common features or some special feature categories to improve this classification. Second, we will study how to filter the general words that negatively affects the classification results now. Third, we also want to evaluate whether the proposed method has the same effectiveness for opinion mining in other language, such as English or Chinese.

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6 Appendix (All data from 2006.09.05)

6.1 Reference words

Hits means the hits from Google using the query “Word (inurl:blog || inurl:bbs || inurl:review || inurl:amazon.co.jp)”. The total number of Weblog pages is 94,500,000, calculated by taking the hits of “は (inurl:blog || inurl:bbs ||inurl:review || inurl:amazon.co.jp)”. “は” means “is” and almost every Japanese page includes the word “は”. Thus the number of pages that include “は” could be considered as the whole Weblog.

The ”R” column of the following tables is the ratio in % of the whole Weblog that contains the word.

Table 6: Positive reference words

Positive Word	Hits (K)	R(%)
いい(good)	44,200	46.77
好き(like)	32,000	33.86
良い(good)	20,500	21.69
大好き(favorite)	14,600	15.45
楽しい(delightful)	14,400	15.24
良かった(good)	11,500	12.17
嬉しい(happy)	10,900	11.53
素敵(lovely)	10,600	11.22
面白い(interesting)	9,880	10.46
魅力(charm)	9,620	10.18
良く(good)	8,880	9.40
欲しい(want)	7,990	8.46
よい(good)	7,290	7.71
満足(satisfaction)	6,810	7.21
素晴らしい(wonderful)	5,270	5.58
おもしろい(interesting)	4,750	5.03
うれしい(happy)	4,730	5.01
すばらしい(wonderful)	3,350	3.54
いいかも(maybe good)	3,300	3.49
お薦め(recommend)	3,290	3.48
優秀(excellence)	3,250	3.44
傑作(masterpiece)	3,180	3.37
悪くない(no bad)	2,140	2.26
旨い(good)	1,540	1.63
優れて(excel)	1,170	1.24
利点(advantage)	809	0.86
嫌いじゃない(not dislike)	635	0.67
気に入り(like)	330	0.35
満たしている(satisfaction)	136	0.14

Table 7: Negative reference words

Negative Word	Hits (K)	R(%)
あまり(not good)	15600	16.51
悪い(bad)	9780	10.35
難しい(hard)	9690	10.25
嫌(dislike)	7580	8.02
不安(worry)	7390	7.82
嫌い(dislike)	6480	6.86
怖い(fearful)	5840	6.18
苦手(hard)	5790	6.13
あんまり(not good)	5490	5.81
不良(bad)	5310	5.62
まずい(bad)	4820	5.10
悪く(bad)	4380	4.63
不満(dissatisfaction)	3760	3.98
最悪(worst)	3620	3.83
危険(risk)	3470	3.67
だめ(useless)	3440	3.64
悲しい(sad)	3380	3.58
辛い(painful)	3200	3.39
不快(dissatisfaction)	2920	3.09
恐ろしい(terrible)	2800	2.96
良くない(bad)	2640	2.79
よくない(bad)	2300	2.43
不具合 (fault)	2190	2.32
面白くない(no interesting)	1490	1.58
欠点(fault)	1430	1.51
悪質(vicious)	1140	1.21
許せない(disallow)	901	0.95
好きではない(dislike)	845	0.89
気に入らない(dislike)	825	0.87
欠陥(defect)	696	0.74
認めない(unacceptable)	470	0.50
満たしていない(dissatisfaction)	44	0.05

6.2 Negated expressions

Table 8: Negated expressions experiment

AC	SC	Negated Expression	SO
P	P	好き(like)	0.82
N	N	好きではない(dislike)	−0.40
N	N	好きではありません(dislike)	−0.34
N	N	悪い(bad)	−0.64
P	(N)	悪くない(not bad)	−0.24
N	N	悪くなくもない(bad)	−0.15
P	P	満足(satisfaction)	0.31
N	N	満足ではありません(dissatisfaction)	−0.40
N	N	満足じゃない(dissatisfaction)	−0.20
P	P	面白い(interesting)	0.16
N	N	面白くない(not interesting)	−0.31
P	(N)	利点(advantage)	−0.06
N	N	利点ではない(disadvantage)	−0.10
N	N	利点ではありません(disadvantage)	−0.17
P	P	良い(good)	0.46
N	(P)	良くない(bad)	0.04
N	N	あまり良くない(not so good)	−0.39
N	N	欠点(fault)	−0.47
P	(N)	欠点ではない(no fault)	−0.13
P	(N)	欠点ではありません(no fault)	−0.10
N	N	許せない(unforgiving)	−0.12
P	P	気に入り(like)	0.25
P	P	気に入っている(like)	0.07
N	N	気に入らない(dislike)	−0.12
N	N	嫌い(dislike)	−1.07
N	N	嫌いじゃないけど(not dislike)	−0.37

AC: [Actual Class] SC: [System Classification]
P: [Positive] N: [Negative]
(): [System assigned the wrong SO value]